Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI / ML engineer shipping production healthcare AI end to end: HIPAA-conscious RAG (~35% retrieval precision, >90% grounded alignment), agentic LLM workflows w/ schema-validated JSON contracts (>30% hallucination reduction), predictive ML pipelines (15-20% recall on high-risk patient categories at >90% precision), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Production computer vision via YOLOv8 (~30% inference latency reduction). Master's in Computer Science, Kent State. Targeting Codes Health's Founding Engineer (NYC Dumbo, healthcare medical-record retrieval YC S24) - direct domain match: HIPAA-conscious RAG over clinical knowledge (~35% retrieval precision, >90% grounded alignment) maps 1:1 to Codes' LLM extraction over patient medical records; Agentic Healthcare Claims 5-stage agent pipeline w/ schema-validated JSON contracts mirrors Codes' "improving LLM pipelines" priority; Manga Lens shipped solo to Chrome Web Store (TypeScript + Manifest V3 + multi-provider) is founding-tier shipping evidence; Dream Decoder full-stack FastAPI + React/TypeScript stack matches Codes' TS/React + Python/FastAPI exactly; EHR preprocessing (Pandas/NumPy, dataset reliability >98%, instability -40%) aligns w/ Codes' record-extraction scope. Open to NYC relocation; F-1 OPT (sponsorship welcome).

CORE COMPETENCIES

HIPAA-Conscious RAG • LLM Extraction over Medical Records • Schema-Validated JSON Agent Contracts • TypeScript + React Frontend • Python + FastAPI Backend • Eval Pipelines + Grounding • Solo-Shipping (Manga Lens, Chrome Web Store) • Multi-Provider LLM Integration • EHR Preprocessing + Data Reliability • Open to NYC Relocation; F-1 OPT

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - Applied AI

USA - Healthcare Technology, Applied AI, SaaS

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, grounding rules; agent response stability **~25%**; hallucinations **>30%**.
- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision **~35%**; retrieval-grounded response alignment **>90%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement scoring, support prioritization; recall **+15-20%** on high-risk categories, precision held **>90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects **~30%**.
- EHR + appointment + ticket preprocessing (Pandas, NumPy); dataset reliability **>98%**, downstream instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; query execution time **~20%**, data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; deployment errors **>30%**, release cycle **~35-40%**.
- Designed **SQL Server DMVs and Grafana** performance dashboards for long-running queries, deadlocks, CPU/IO contention; incident recurrence **~25%**, RCA speed **~30%**.
- Tuned database maintenance (index rebuilds, partition-aware indexing); reconciliation runtime **~15%**, deadlocks **~15%**.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff (research-to-production handoff).

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake Normalization, Validation, Consistency Analysis, Duplicate Detection, Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts, vector search

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction across **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) with per-provider payload handling for failure isolation. Manifest V3 service workers; 7-day translation cache; cost-aware payload routing (WebP for cloud, JPEG for local).

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal AI APIs (Perplexity Sonar interpretation + Replicate image generation). Intermediate **structured prompt transformation layers** raised contextual alignment **~30%** and first-pass image success **~25-30%** over naive direct-prompt orchestration.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate, multimodal orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with a unified **YOLOv8** detection-and-classification model - inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping + NMS tuning for stable real-time operation via OpenCV video capture.

PyTorch, YOLOv8, OpenCV, LabelImg, real-time inference

EDUCATION

Master of Science, Computer Science - *Kent State University* *Aug 2023 - May 2025*

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - *KL University* *Jun 2019 - May 2023*

Founder / Lead of E-Farming platform (university project).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI (Coursera)* *2024*

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++

LLM / GenAI: RAG, agent workflows (memory, tool use, multi-step), structured outputs, evaluation pipelines, embeddings, vector search, LangChain, LlamaIndex, multi-provider LLM integration (OpenAI / Anthropic / Gemini / open-source), schema-validated JSON agent contracts

ML Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, threshold calibration, walk-forward validation

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose / MediaPipe, Stable Diffusion, video analytics

Backend / Inference: FastAPI, Flask, REST, Docker, structured logging, load simulation

Frontend: React, TypeScript, Vite, Tailwind, Chrome MV3 service workers

AI-First Dev: Cursor, Claude Code, Copilot, AI-assisted code validation

Data: PostgreSQL + pgvector, Pandas, NumPy, EHR preprocessing, time-series

DevOps: Docker, Jenkins, GitHub Actions, Grafana, cloud-ready deployment

Domains: Applied AI, Healthcare AI, HIPAA-Conscious, Agentic / Multi-Step LLM, Production RAG, Multimodal Systems